

Namespaces and Vocabularies

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Overview

Namespaces provide short, readable prefixes for long URI bases. RDFLib.jl comes with many predefined namespaces for common RDF vocabularies and lets you create your own.

```
using RDFLib
```

Predefined Namespaces

RDFLib.jl includes all major W3C and community vocabularies:

```
# Core W3C vocabularies
println("RDF type:           ", RDF.type)
println("RDFS label:         ", RDFS("label"))
println("OWL Class:           ", OWL("Class"))
println("XSD integer:          ", XSD.integer)
println("XSD dateTime:         ", XSD.dateTime)

# Community vocabularies
println("FOAF name:           ", FOAF("name"))
println("DC title:             ", DC("title"))
println("DCTERMS created:      ", DCTERMS("created"))
println("SKOS prefLabel:       ", SKOS("prefLabel"))
println("PROV wasGeneratedBy: ", PROV("wasGeneratedBy"))
println("Schema.org name:      ", SDO("name"))
```

RDF type:	URIRef("http://www.w3.org/1999/02/22-rdf-syntax-ns#type")
RDFS label:	URIRef("http://www.w3.org/2000/01/rdf-schema#label")
OWL Class:	URIRef("http://www.w3.org/2002/07/owl#Class")
XSD integer:	URIRef("http://www.w3.org/2001/XMLSchema#integer")

```
XSD dateTime:      URIRef("http://www.w3.org/2001/XMLSchema#dateTime")
FOAF name:         URIRef("http://xmlns.com/foaf/0.1/name")
DC title:          URIRef("http://purl.org/dc/elements/1.1/title")
DCTERMS created:   URIRef("http://purl.org/dc/terms/created")
SKOS prefLabel:    URIRef("http://www.w3.org/2004/02/skos/core#prefLabel")
PROV wasGeneratedBy: URIRef("http://www.w3.org/ns/prov#wasGeneratedBy")
Schema.org name:   URIRef("https://schema.org/name")
```

Creating Custom Namespaces

```
# Simple namespace from a URI base
bio = Namespace("http://purl.org/vocab/bio/0.1/")
println(bio("Birth"))
println(bio("Event"))

# Use namespaces to build URIs concisely
dbr = Namespace("http://dbpedia.org/resource/")
dbo = Namespace("http://dbpedia.org/ontology/")

g = RDFGraph()
add!(g, Triple(dbr("Julia_Language"), RDF.type, dbo("ProgrammingLanguage")))
add!(g, Triple(dbr("Julia_Language"), dbo("designer"), dbr("Jeff_Bezanson")))
add!(g, Triple(dbr("Julia_Language"), RDFS("label"), Literal("Julia"; lang="en")))
```

```
URIRef("http://purl.org/vocab/bio/0.1/Birth")
URIRef("http://purl.org/vocab/bio/0.1/Event")
```

```
RDFGraph (3 triples)
```

Binding Prefixes to Graphs

Prefix bindings control how URIs are shortened in serialized output:

```
bind!(g, "dbr", Namespace("http://dbpedia.org/resource/"))
bind!(g, "dbo", Namespace("http://dbpedia.org/ontology/"))

println(serialize(g, TurtleFormat()))
```

```

@prefix dbo: <http://dbpedia.org/ontology/> .
@prefix dbr: <http://dbpedia.org/resource/> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

dbr:Julia_Language a dbo:ProgrammingLanguage ;
    dbo:designer dbr:Jeff_Bezanson ;
    rdfs:label "Julia"@en .

```

Namespace Manager

The `NamespaceManager` provides centralized prefix management:

```

nsm = NamespaceManager()
bind!(nsm, "ex", Namespace("http://example.org/"))
bind!(nsm, "foaf", Namespace("http://xmlns.com/foaf/0.1/"))
bind!(nsm, "schema", Namespace("http://schema.org/"))

# Expand CURIEs (Compact URIs)
println(expand_curie(nsm, "ex:alice"))
println(expand_curie(nsm, "foaf:name"))

URIRef("http://example.org/alice")
URIRef("http://xmlns.com/foaf/0.1/name")

# Compute qualified names (reverse of expand)
qname = compute_qname(nsm, URIRef("http://xmlns.com/foaf/0.1/Person"))
println("QName: ", qname)

QName: ("foaf", "http://xmlns.com/foaf/0.1/", "Person")

```

Programmatic Namespace Creation

Create new vocabulary namespaces with metadata:

```

# create_namespace generates Julia code for a DefinedNamespace from a graph
# First, build a small ontology graph
vocab_g = RDFGraph()
vocab_ns = Namespace("http://example.org/myvocab#")
add!(vocab_g, Triple(vocab_ns("Person"), RDF.type, RDFS("Class")))
add!(vocab_g, Triple(vocab_ns("hasFriend"), RDF.type, RDF("Property")))

code = create_namespace(vocab_g, "http://example.org/myvocab#", "MyVocab")
println(code)

# For direct use, just create a Namespace
my_vocab = Namespace("http://example.org/myvocab#")
println(my_vocab("Person"))
println(my_vocab("hasFriend"))

```

```

# Auto-generated namespace: MyVocab
# URI: http://example.org/myvocab#

```

```

const MyVocab = DefinedNamespace(
    "http://example.org/myvocab#",
    Set([
        "Person",
        "hasFriend"
    ])
)

URIRef("http://example.org/myvocab#Person")
URIRef("http://example.org/myvocab#hasFriend")

```

Using Namespaces in Practice

A common pattern is to define all your namespaces at the top of your code:

```

# Define project namespaces
ex = Namespace("http://example.org/")
schema = Namespace("http://schema.org/")

# Build a well-prefixed graph
g = RDFGraph()
bind!(g, "ex", ex)
bind!(g, "schema", schema)

```

```

bind!(g, "xsd", Namespace("http://www.w3.org/2001/XMLSchema#"))

# Describe a conference
conf = ex("juliacon2025")
add!(g, Triple(conf, RDF.type, schema("Event")))
add!(g, Triple(conf, schema("name"), Literal("JuliaCon 2025")))
add!(g, Triple(conf, schema("startDate"), Literal("2025-07-21"; datatype=XSD.date)))
add!(g, Triple(conf, schema("endDate"), Literal("2025-07-25"; datatype=XSD.date)))
add!(g, Triple(conf, schema("location"), Literal("Aachen, Germany")))
add!(g, Triple(conf, schema("url"), URIRef("https://juliacon.org/2025/")))

println(serialize(g, TurtleFormat()))

```

```

@prefix ex: <http://example.org/> .
@prefix ns1: <https://juliacon.org/2025/> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix schema: <http://schema.org/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

```

```

ex:juliacon2025 a schema:Event ;
    schema:endDate "2025-07-25"^^xsd:date ;
    schema:location "Aachen, Germany" ;
    schema:name "JuliaCon 2025" ;
    schema:startDate "2025-07-21"^^xsd:date ;
    schema:url ns1: .

```

Common Vocabulary Prefixes

Here is a quick reference of commonly used namespace prefixes:

Prefix	URI	Purpose
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#	RDF
rdfs:	http://www.w3.org/2000/01/rdf-schema#	RDF Schema
owl:	http://www.w3.org/2002/07/owl#	OWL
xsd:	http://www.w3.org/2001/XMLSchema#	XML Schema datatypes

Prefix	URI	Purpose
foaf:	http://xmlns.com/foaf/0.1/	Friend of a Friend
dc:	http://purl.org/dc/Elements/1.1/	Dublin Core
dcterms:	http://purl.org/dc/terminology/	Dublin Core Terms
skos:	http://www.w3.org/2004/02/skos/core/#	SKOS Core
schema:	http://schema.org/	Schema.org
prov:	http://www.w3.org/ns/prov#	Provenance
shacl:	http://www.w3.org/ns/shacl#	SHACL Constraint Language